

The Cirqen Ansur Connector

A plan for reading every Ansur test into Cirqen automatically, in the background, with the device identified by its serial number, the test filed under a named Ansur procedure for that equipment description, every measurement summarised, and the measurement uncertainty that Ansur does not calculate added by Cirqen. Manual keying stays available for tests done without Ansur, in the same form.

Prepared	24 September 2026, for the hospital biomedical engineering department and the Cirqen team
Status	Plan. Nothing in this document is built yet. The hospital has not yet supplied an Ansur result file, so the file format is an open question with its own discovery phase (§8).
Builds on	Cirqen branch <i>checklists-warranty-notifications</i> : checklists, email alerts, failure risk, and the result-import design used for MET/CAL, which Ansur reuses.
Overall rating	7.5 / 10 as a plan today, rising to about 8.5 once real result files confirm the format (§11). Readiness of the running system for Ansur today: 1.5 / 10.

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1 The situation

The hospital tests medical equipment with Fluke Biomedical analysers driven by **Ansur 3.1.4** test-automation software. Ansur is a Fluke *Biomedical* product for safety and performance testing (electrical safety analysers, patient simulators, infusion pump and defibrillator analysers). It is not Fluke Calibration's MET/CAL, so MET/CONNECT is not needed and nothing has to be bought to connect it.

What Fluke Biomedical publishes about Ansur, and what it means for Cirqen:

Published fact	Consequence for the connector
Each test is saved as a result file (.mtr, XML) ; a report can be generated as PDF .	The primary source is the result file. The PDF is a secondary source used to cross-check and as a fallback (§3.4).
Results can be exported from the analyser as CSV .	A third source for tests run on the analyser alone, without Ansur (§7).
The Ansur 3 datasheet lists Microsoft SQL Server 2012 among its requirements.	Some installations keep results in a database. The detector checks for it; if present, the connector reads it with a read-only login instead of the folder.
A test record carries asset ID, serial number and a work order reference.	Enough to match every result to a Cirqen device and work order without anyone re-typing.
Licences are per analyser plug-in, per PC. Fluke offers an interface developer's kit .	Reading the hospital's own result files needs no extra licence. The kit is requested in discovery.
Ansur records the reading and a pass/fail against the template limit, not measurement uncertainty.	Cirqen computes uncertainty, test uncertainty ratio and a guard-banded verdict itself (§6).

The one unknown. Fluke does not publish the layout of the .mtr file, and no sample has been supplied yet. The plan is designed so that everything except the parser can be built now, and the parser is built and proven against real files in a short discovery phase (§8).

2 What the connector does, end to end

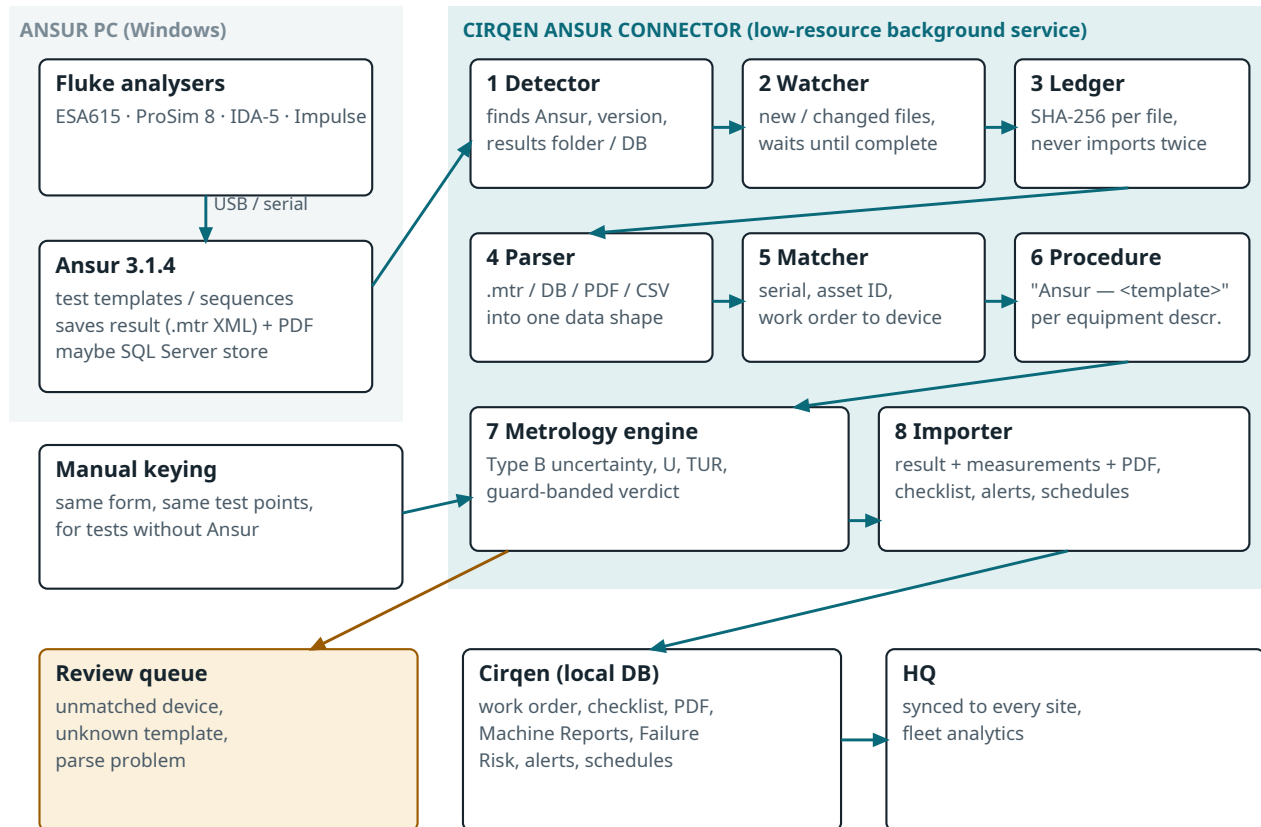


Figure 1. Data flow. The connector only reads Ansur's results and never writes to Ansur. Orange: results it cannot match go to the review queue. Results reach HQ through normal sync.

A day in the life:

1. The technician opens Ansur, picks the template for the device (for example a patient monitor PPM sequence using the ESA615 and ProSim 8), scans the device's **asset barcode** and types or scans the **Cirqen work order reference**, runs the test and presses Save, exactly as today.
2. Within about a minute the connector notices the new result, waits until Ansur has finished writing it, fingerprints it, and reads it.
3. It finds the device by **serial number** (and asset ID), attaches the result to the open work order, and files it under the procedure "**Ansur — <template name>**" for that equipment description.
4. Every measurement is stored with its unit, limit, condition and Ansur's own pass/fail. Cirqen adds the uncertainty, TUR, margin to the limit and its own guard-banded verdict.
5. The work order's checklist steps that Ansur covered (for example "Electrical safety test (IEC 62353)") are filled in automatically; the visual steps are still answered by the technician.
6. A failure or an indeterminate result emails the department In-Charge, copying the HOD. A passing PPM with a complete checklist closes the PPM schedule. Everything appears in Machine Reports and syncs to HQ.

3 The connector, part by part

3.1 Detector

Runs at start-up and once an hour. It answers: is Ansur on this PC or reachable from it, which version, where are the results, and which analyser plug-ins are installed.

- **Installation:** Windows uninstall entries and the Fluke Biomedical program folder give the Ansur version (3.1.4) and plug-in list. A version the parser has not been proven on is flagged, not guessed at.
- **Results location:** Ansur's own settings (its configured save folder), then the usual document folders, then a path set by an administrator. A network share works the same way, so a Cirqen desktop on Linux can serve an Ansur PC over the network.
- **Database:** if a SQL Server instance with an Ansur database is present, it is reported and, with a read-only login from hospital IT, used in place of the folder.
- **Status page** in Cirqen (System Ansur Connection): detected version, source, last file seen, files imported today, waiting for review, and the last error in plain words.

3.2 Watcher

- Uses operating-system file notifications where available, with a **60-second polling fallback** for network shares, where notifications are unreliable.
- A file counts as complete only when its size and modification time have been stable for 5 seconds and it opens without a lock, so a half-written file is never read.
- Skips Ansur's templates, temporary files and anything older than a configurable start date, so the first run does not flood Cirqen with years of history unless asked to (a separate back-fill command does that, at a gentle rate).

3.3 Ledger: exactly once

- Each file is fingerprinted (SHA-256 of its bytes). The ledger records the fingerprint, path, result, and outcome. The same bytes are never imported twice; a re-saved, changed file updates the same Cirqen result and keeps the previous version in history.
- The raw file is stored compressed with the result, so a parser improvement can re-read every past result without going back to the Ansur PC.
- The Cirqen result ID is derived from Ansur's own test identifier (or the fingerprint), so two desktops that read the same file converge on one record through sync, as the MET/CAL import already does.

3.4 Parser

- **Four readers, one output.** .mtr (XML) is primary; the SQL Server store if present; the PDF report (text extraction) as a fallback; analyser CSV for standalone tests. All produce the same normalised structure, so everything downstream is written once.
- **Versioned.** The parser records which Ansur version and template it understood. Unknown elements are kept as raw data, not dropped; an unfamiliar file goes to the review queue with the reason.
- **Self-checking.** When both the .mtr and its PDF exist, the parser compares the values it read from each. A mismatch blocks the automatic import and asks a person to look. During the pilot this checks every file; afterwards a sample.

3.5 Matcher

Order	Match on	Notes
1	Serial number	Normalised (upper case, spaces and dashes removed). Required to be recorded on every result. The serial is stored even when another key made the match.
2	Hospital asset number	Same as the barcode on the device; best when scanned in Ansur.
3	Cirqen work order reference	Links the result to the work order; if blank, the device's open work order from the same day is used.
—	No match	Review queue: the technician picks the device once; the fix is remembered for that serial spelling.

3.6 Low-resource by design

Budget	Target	How
CPU when idle	< 1 %	Sleeps between checks; one directory scan per minute at most.
Memory	< 40 MB	Streams each file; no caches of past results.
Disk	~ 30 KB per test	Raw file compressed; PDFs stored once.
Network	Only for sync	Parsing and matching are local; results go to HQ with normal sync.
Fairness	Backs off	Pauses while CPU is above 80 % or the PC is on battery; never slows a test.

It runs inside the Cirqen background worker that already sends email and syncs, so there is no new program to install or keep alive. If Cirqen is switched off, it catches up from the ledger when it starts.

4 Procedures: labelling Ansur tests per equipment description

Every Ansur template or sequence becomes a Cirqen procedure the first time it is seen, named "**Ansur — <template name>**", marked with source *Ansur*, and linked to the equipment description of the device it was run on (the existing *EquipmentCalibrationProcedure* link). The procedure records:

- the template name and a fingerprint of its test steps (a changed template becomes a new version, so old results keep the steps they were run with);
- the analysers it uses (for example ESA615 + ProSim 8);
- the list of test points, with units, limits and the standard they come from (IEC 62353 or IEC 60601-1);
- which checklist steps it satisfies (set once in Checklist Settings, for example "Electrical safety test" and "NIBP pressure accuracy").

A new procedure is created as **provisional** and appears in the review queue: a senior technician confirms the equipment description and checklist mapping once. After that, every result on that template files itself.

Equipment description	Example Ansur procedure	Test points it contributes
Patient Monitor	Ansur — PM PPM (ESA615 + ProSim 8)	Earth resistance, enclosure and patient leakage (normal and single fault), NIBP accuracy at set pressures, ECG rate, SpO2 at set saturations, alarms
Infusion Pump	Ansur — Pump PPM (ESA615 + IDA-5)	Electrical safety, flow rate accuracy, occlusion pressure, bolus volume
Defibrillator	Ansur — Defib PPM (ESA615 + Impulse 7000)	Electrical safety, delivered energy at set levels, charge time, synchronised delay

The names above are examples of the pattern; the real names come from the hospital's Ansur templates.

5 Measurements: the full summary

Every test point Ansur records is stored and shown, not only the verdict. For each point:

Field	From	Example
Group	Ansur	Electrical safety · NIBP · ECG · SpO2 · Flow · Energy
Test point and condition	Ansur	Patient leakage, applied part BF, single fault: open neutral
Reading and unit	Ansur	0.112 Ω
Limit and its source	Ansur template; checked against Cirqen's IEC limit library (§9)	0.300 Ω (IEC 62353)
Ansur verdict	Ansur	Pass
Expanded uncertainty U, k	Cirqen (§6)	0.0205 Ω, k = 2
Test uncertainty ratio	Cirqen	14.6 : 1
Margin to limit	Cirqen	63 % of the limit unused
Cirqen verdict	Cirqen (guard-banded)	Pass / Fail / Indeterminate
Analyser and its calibration	Ansur + Cirqen standards register	ESA615 SN ..., calibrated to 12/2026

The work order and its PDF get a **test summary**: counts per group, every failed or indeterminate point at the top, the smallest margins, and the full table after. Machine Reports gets the same data per device over time, so a leakage current creeping upward over several PPMs is visible before it fails (\$9).

6 Uncertainty: what Cirqen adds that Ansur does not

Ansur usually records **one reading per test point**. With one reading there is no scatter to measure, so the uncertainty comes from what is known about the measuring chain (GUM Type B), with scatter added from history once enough tests exist. Cirqen already has the engine for this in CalSoft (uncertainty budget, Welch–Satterthwaite degrees of freedom, coverage factor, TUR, guard-banded decision and drift analysis); the connector feeds it.

Component	Source	Standard uncertainty
Analyser accuracy	Analyser datasheet, per function and range, entered once per model in an analyser specification library	$u = a / \sqrt{3}$ (rectangular)
Display resolution	Analyser range	$u = r / \sqrt{12}$
Analyser's own calibration	The analyser registered as a standard in CalSoft, with its certificate	$u = U_{\text{cert}} / k_{\text{cert}}$
Repeatability (later)	Pooled standard deviation of repeat tests on the same model and test point, once 10 or more exist	$u = s_{\text{pooled}}$ (Type A)

Combined and expanded:

$u_c = \sqrt{(\sum u_i^2)}$ $U = k \cdot u_c$ $k = 2$, or from the t-distribution when repeatability is included (effective degrees of freedom)

Decision (ILAC-G8 guard banding, the rule already in CalSoft), for a one-sided limit L such as a leakage current or earth resistance:

reading $\leq L - U \rightarrow$ PASS
 reading $\geq L + U \rightarrow$ FAIL
 otherwise $\rightarrow$ INDETERMINATE (retest, or an accepted-risk decision by an authorised person, recorded)

Worked example: protective earth resistance

Illustrative numbers. The analyser specification below is an example chosen to show the method. The real values come from the hospital analysers' datasheets and calibration certificates in phase 3.

Quantity	Value	Working
Limit L	0.300 Ω	IEC 62353, protective earth with detachable power cord
Reading	0.112 Ω	From the Ansur result
Analyser accuracy a	0.0172 Ω	$\pm(2\% \text{ of reading} + 0.015 \Omega) = 0.02 \times 0.112 + 0.015$
u_1 accuracy	0.00995 Ω	$0.0172 / \sqrt{3}$
u_2 resolution	0.00029 Ω	$0.001 / \sqrt{12}$
u_3 analyser calibration	0.00250 Ω	$0.005 / 2$
u_c	0.01027 Ω	$\sqrt{(u_1^2 + u_2^2 + u_3^2)}$
U (k = 2)	0.0205 Ω	2×0.01027
Acceptance limit	0.2795 Ω	$L - U = 0.300 - 0.0205$
TUR	14.6 : 1	L / U ; 4 : 1 or better is the usual target
Verdict	PASS	$0.112 \leq 0.2795$; 63 % of the limit unused

Engineering change needed: CalSoft's uncertainty budget currently needs at least two readings. Phase 3 adds a single-reading path (Type B only) and the analyser specification library; nothing else in the engine changes.

7 Manual keying and Ansur side by side

Ansur will not cover every test: an analyser used on its own, a device type with no Ansur template, a test done on a ward, or an Ansur PC that is down. The plan keeps one result structure with three ways in:

Way in	When	How it works
Ansur (automatic)	Normal case	Connector, as above. No typing.
Manual keying	No Ansur for this test	A form built from the same procedure's test points (learned from Ansur files, or written by hand for devices Ansur never tests). The technician picks the analyser from the standards register and types readings; limits, uncertainty and verdict are filled in the same way.
Analyser CSV	Analyser used without Ansur	Upload or drop the CSV exported from the analyser memory; mapped onto the same test points.

- Every result shows its **source** (Ansur, Manual, CSV), and reports treat them the same.
- If a manual entry and an Ansur file arrive for the same work order and test, both are kept and shown side by side for a reviewer; nothing is overwritten silently.
- Manual entries are marked lower-assurance on reports (typed, not captured), which is an honest statement for an auditor and an incentive to use Ansur where possible.
- The work order checklist is the meeting point: visual steps are always answered by the technician; measured steps are filled from Ansur when a result exists and keyed otherwise.

8 Getting the file format without samples

No .mtr file is available today. The plan does not guess the format; it gets it, in this order:

Step	Action	Output
1	Collect 10–20 real result files from the Ansur PC, covering every analyser plug-in in use, at least one failed test, and the PDF report of each.	The golden test set
2	Request Fluke Biomedical's interface developer's kit (techservices@flukebiomedical.com), which documents the result data for CMMS developers.	Official field meanings
3	Check whether Ansur 3.1.4 on that PC stores results in SQL Server; if so, get a read-only login and map the tables.	Folder or database decided
4	Build the parser against the golden set; every file must reproduce the numbers on its PDF exactly.	Parser + automated tests
5	Pilot: run silently for two weeks alongside normal work, comparing every import with its PDF, before turning on automatic filing.	Proven on live data

Fallback if the .mtr format cannot be read reliably: the PDF report reader alone (text extraction of the result table) gives the same measurements with slightly less structure. The connector design does not change; only the reader does.

9 Further advancements

Beyond reading the data, these make Cirqen the place where test quality is checked, not just stored.

Advancement	What it gives	Effort
Analyser traceability	Each ESA615, ProSim, IDA and Impulse is a standard in CalSoft with its certificate and due date. Results taken with an analyser past its due date are flagged and alerted.	Low
IEC limit library	Cirqen holds IEC 62353 / 60601-1 limits by device class (I, II) and applied part (B, BF, CF) and re-checks Ansur's limit. Catches a test run on the wrong template (for example BF limits on a CF device).	Medium
Leakage and resistance trends	The drift analyser already in CalSoft fitted to leakage current and earth resistance per device; rising insulation leakage is an early sign of failure and feeds Failure Risk.	Low
Auto-close PPM	A passing Ansur PPM with a complete checklist closes the PPM schedule and dates the next one.	Low
Barcode everywhere	One asset label scanned in Ansur and in Cirqen removes most matching errors.	Low (labels)
Integrity checks	Flags results that look copied (identical values on different devices), tests shorter than the template's minimum duration, and edits after saving. Raw file fingerprints make tampering visible.	Medium
Fleet analytics at HQ	Failure rate by model and by test point across sites; which models fail leakage most; which analysers disagree.	Medium
Alerts with context	Failure email includes the failing points, their margins, and the device's last three results for the same point.	Low
Result PDF with uncertainty	A Cirqen test report alongside Ansur's own, adding U, TUR and the guard-banded verdict for accreditation (ISO/IEC 17025-style reporting).	Medium
OneQA (optional, later)	Fluke Biomedical's newer platform has built-in CMMS links. Not needed for this plan; worth revisiting only if the hospital moves to it.	—

10 Delivery phases

Phase	Scope	Needs	Length	Done when
0 Discovery	Golden files, developer kit, folder or DB decided	Files from the hospital	1 week	Format understood
1 Connector core	Detector, watcher, ledger, raw storage, status page, review queue	Nothing external	1–2 weeks	Files detected and fingerprinted on the Ansur PC
2 Parser and matching	Readers, normalised model, matcher, procedure labelling	Phase 0	2 weeks	Golden set parses; PDF cross-check passes
3 Metrology	Single-reading uncertainty, analyser spec library, guard-banded verdict, IEC limit library	Analyser datasheets and certificates	1–2 weeks	Worked examples reproduced by hand
4 Manual and CSV	Keying form from procedure test points, analyser CSV import, source marking	Phase 2	1–2 weeks	Same result shape from all three sources
5 Outputs	Work order summary and PDF, checklist fill, alerts, PPM auto-close, Machine Reports, trends	Phases 2–4	1–2 weeks	A PPM runs end to end with no typing
6 Pilot	One workshop, silent then live; every import compared with its PDF	Technician time	4 weeks	200 results with zero unexplained differences

About 11–15 weeks end to end. Phases 1 and 3 can start now; only phase 2 waits for the files.

11 Ratings

Out of 10. "Today" is the running system for Ansur data (nothing is connected, so results are PDFs on a PC). "After plan" is the expected level once phases 0–6 are complete. The bar shows today in grey and the plan in teal; the black tick is the target.

Aspect	Today	After plan		Main limit
Automatic capture (no manual uploads)	0.5	9.0		File watch or DB read; only Save in Ansur remains.
Device and work order matching	2.0	8.5		Depends on asset ID discipline; review queue for the rest.
Measurement detail in Cirqen	0.5	9.0		Every point with limit, condition and verdict.
Uncertainty and decision rule	0.0	8.0		Type B only at first; as good as the analyser specs entered.
Procedure labelling per description	1.0	8.5		Automatic, with one-time confirmation per template.
Manual keying alongside Ansur	3.0	8.5		Same structure; source marked; conflicts reviewed.
Traceability of analysers	2.0	8.5		Analysers as standards with due dates.
Background footprint	0.0	9.0		Inside the existing worker; strict CPU and memory budget.
Resilience (offline, restarts, upgrades)	0.0	8.0		Ledger and raw files; Ansur upgrades need a parser check.
Confidence in the plan itself	6.0	8.5		Held back only by the unseen .mtr format.
Mean	1.5	8.6		Plan rating today: 7.5 (a sound design with one unknown input)

12 Risks

Risk	Effect	Mitigation
.mtr format harder than expected	Phase 2 slips	PDF reader fallback; developer kit; raw files kept for re-parse.
Ansur upgrade changes the format	New results stop importing	Detector notices the new version; unfamiliar files go to review, never half-imported; golden tests extended.
Asset IDs not entered in Ansur	Many results in review	Barcode labels and scanners; the matcher learns corrections; weekly count shown to the HOD.
Analyser specifications unknown	Uncertainty not computable	Verdict falls back to Ansur's own and says so; specs collected in phase 3.
Ansur PC is Windows, Cirqen desktop may be Linux	Connector cannot run locally	Cirqen's Windows build on the Ansur PC, or a read-only network share to a nearby Cirqen desktop.

Risk	Effect	Mitigation
Reading a vendor file format	Support or licence questions	Read-only access to the hospital's own data; developer kit requested; no Ansur files modified.

13 What we need from the hospital

- 10–20 Ansur result files (.mtr) with their PDF reports, covering each analyser in use and at least one failure.
- Where Ansur saves results on the Ansur PC, and whether an SQL Server database is used.
- Whether a Cirqen desktop can run on the Ansur PC, or a read-only network share to it is acceptable.
- The list of analysers (model, serial number) with their latest calibration certificates and datasheets.
- Agreement on the decision rule (guard banding, and who may accept an indeterminate result).
- Barcode asset labels on devices, or agreement to use the existing asset numbers in Ansur's asset ID field.

Sources: Fluke Biomedical, *Ansur Test Automation* product page; *Ansur Test Automation Software FAQs* (9460563); *Ansur 3.0 Technical Data*; *OneQA* product page. Cirqen CalSoft module (CalSoft/utlis.py) for the uncertainty and decision methods.